

Basics of Active Inference (Discussion) ~ Ben White

~ Active Inference for the Social Sciences 2023

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all right July 25th 2023 and we are in the discussion section basics of active inference this is the first discussion section that we've had for this course so thank you all for joining and it's going to be regarding the topics that Ben White introduced in his recent lecture so everyone will be welcome to pop in and introduce themselves and then I know that Ben has some ideas to discuss and many other spontaneous and written questions will come into play so I will uh with that exit for now and pass to perhaps some of our first time live stream guests I guess I've already spoken so uh I may as well continue so I'm Darius um I am Master student I'm just about to finish at UCL and sort of social distributed cognition I work in the social cognition lab looking at salience regulation and attentional mechanisms within social contexts and but all within an active inference framework um within the Bayesian brain hypothesis framework and yeah it's a sort of recent Discovery and Obsession and so I'm sort of going to grips with the both the high road and the low road um and so I've had the opportunity to chat to Mark and some other researchers who have been super informative but always looking to learn more and get my head around even more of the sort of philosophical and mathematical Theory oh sounds good so hey everyone can you hear me yep I'm Francisco balzan I'm a PhD student at the University of Bologna Italy actually really working on the intersection between artificial intelligence and education and I have a background in cognitive anthropology and philosophy of science and I'm at excellent I'm actually in a few years ago so I stepped out into the rabbit program active inference and currently pretty much interested in uh um multi-scale active inference models are the scientific technicians so pretty interested about the agent level modeling of scientific reasoning the assumption that might emerge from the interaction with the scientific environment so we're referring to Scientific music construction and all this type of top-down environment interactions so super interested to do something from you thank you thank you very much would anybody else like to introduce themselves before we start thank

you so you can first okay hi hi I'm Regina Sagi umina I'm from Guatemala and well I'm not from the social sciences I come from biology and Neuroscience and um I'm here because um I'm working on a project as a technical operator in this big project in DX Escape material mines and I do the technical part and experiments but I want to learn the philosophical part that's the background that you know makes The Theory of Everything we're going to you know experiment so I'm here to learn um and also I am like you know when you see the Olympics that you you see all the swimmers and you always say I wish there was a normal guy to see how good these people are so you could compare so that's me today right I I understand half of what's going on but I'm happy to be here so yeah hi oh yeah sorry I'm late um sorry my name's Lee um I'm a PhD student at the uh University of New York um studying systems transformation um and I'm I'm joining because I'm using uh perceptual control theory at the minute to uh to model examples of effective practice um in organizations and and I've read quite a lot of um active inference papers and I understand that there's a lot of resonance uh and overlap between perceptual control theory and um active influence although I understand it's within a predicted framework and also it's uh it's a kind of a level of abstraction higher so you're able to kind of quantify the difference between um sort of the the predictive State or the outcome State and you know in the current state so what I'm really trying to understand is how might that level of abstraction be useful um in what I'm doing actually because I understand it's a you know a much more kind of concurrent Theory and framework than um then perceptual control theory yeah very cool um anybody else I think there's was that everybody okay well um was everybody um present at the has everybody seen the lecture that I gave a couple of weeks ago okay so what I think it might be helpful then if I kind of very very briefly go over what what we covered in that lecture just to kind of jog people's memories and then we can maybe pick up some questions from there um so the lecture last week was on the basics of active inference and it was um intended to be an introduction to the framework on a very abstract philosophical level so it left out all of the kind of mathematics and Technical aspects of the framework and the objective really was to lay the groundwork for future weeks in this course because um this is obviously a course on active inference in the social sciences and so in the subsequent weeks we're going to be looking at topics like Collective Behavior social cognition uh shared norms and Niche construction and so what I wanted to do was to put on the table a kind of fully fleshed out picture of what the individual agent looks like in active inference and so to do this um we looked at uh emotion agency uh mind so we looked at kind of the role of action um and perception um the role of internal representations um

inactive imprints and then as a case study we kind of we looked at a active inference based account of addiction to kind of bring all of these threads together um to kind of articulate a lot of the things that that I'd said in the previous sections so I'm I mean the questions here don't have to be specifically um Tethered to that lecture I suppose we can just start with any general questions about active inference and philosophy and how active inference applies to individual agents or if anybody did want to pick up on any threads from the lecture then we can go from there can I ask something just for the basic questions yeah of course um I don't understand because I think you explain it later but how how does creativity and science in general enters active imprints how does trying to be creative and not put doing what you predict enters this like role yeah yeah this is a really good question and I think um so were you there have you did you see the lecture yeah but I remember half of it yeah that's that's fine but I think uh I think do you remember the dark room problem yeah yeah I know that's that's where the answer is but yeah well that's it's it's it's I think there are several interested in that dark room problem some of them are some of them more interesting than others and I I tried to cover I did obviously didn't have time to cover all of those in the lecture um but the kind of for those that perhaps aren't familiar the dark room problem is um this kind of canonical philosophical worry um about predictive processing and active inference that says essentially if it's if all we're trying to do is minimize prediction errors why don't we um just find maximally predictable environments and you know why do we not just lay in a dark room hooked up to an intravenous drip and enjoy a kind of error-free Lifestyle um so I think that's the kind of that's the question that you're articulating there um I just quickly see a hand from Darius Darius did you want to jump in or is it a separate question it's a separate question so I would just do it in advance but yeah no ignore me I'll Circle back around in a second then you're free um I should say actually um this is the first time I've kind of shared a discussion like this so I'll be explicit and say um people should just feel free to jump in whenever they want if you want to take the Disco If you want to kind of um add something or or kind of build on a question that we're answering feel free to jump in um but yeah with it with a dark room problem the as I as I'm aware the first answer to that question and this is something that I did Cover in the lecture is essentially a big part of active inference is to recognize that the agents are it's a kind of fundamentally embodied framework okay so what that that can be kind of cashed out in various ways but one of the one of the most important ways is that the the expected states of the organism or that the agent are necessarily rooted in the phenotype of the organism and the the kind of evolutionary biological needs that come with having a particular

body so there was a paper that I referenced in a lecture by that's three authors on this paper it's Andy Clark call for assistant and then I'm forgetting the name of the third author if anybody can recall um feel free to drop it in the chat but this is this is an answer to the dark room where it builds on that notion of embodiment and says look um creatures like us humanism in the dark room and do nothing and just uh kind of enjoy the very predictable March of hunger Thursday Etc but obviously those particular biology of the room and so you have this very fundamental level we have this very fundamental um kind of basis upon which we have these needs that need to be met and so we need to engage in exploratory behaviors um on the topic of novelty um because that's because that's the kind of Base I take that to be a kind of Baseline answer um but there are clearly um kinds of activities and behaviors that we do engage in um going beyond the dark room kind of curiosity play um creativity engaging in kind of why do we create works of art and it there's not an obvious link between those kinds of behaviors and the kinds of biological needs um that are emphasized in the original answer to the dark room worry um so in answer to Regina's question about creativity I think that a good starting point for this is the section on aerodynamics that I introduced in the lecture so this was the idea that it there's this strong connection between affectivity as a kind of emotional embodied feeling State and the changes to the rate in prediction error minimization so I don't know um I am now very much uh articulating Mark's Work here so I don't know if Mark wants to jump in at any point and do a much better job than I can yeah I think there's there's certainly there's certainly work on the horizon well I know that there are people who are currently building a series of artistic pretty that are that are based on this this idea of aerodynamics so Mark's paper on play with Mark andon that I mentioned has really interesting idea where if you explain playful behaviors which are kind of fundamentally created and very we are you know it's inherently creative exercise I think the really interesting idea that comes out of that is one thing that we engage in because it feels good is we we create niches of very manageable prediction error that we can then minimize um so yeah that's playful behavior on is the kind of the answer to the puzzle of play so why do we in the puzzle of the players why do we engage in previous uh metabolically quite costly and they don't have any obvious benefit and the idea there's that well there's there's kind of several ideas in there um but one of them is that it just literally feels good because minimizing prediction error feels good to us so whenever we do better than expect at minimizing prediction error it feels good and so I think so obviously I don't know for sure but I think that any answer that we give regarding artistic creativity is going to fall within that kind of bulk that essentially we engage um in the part of the creative process is just

constructing those that give us tasks games of manageable prediction error that ideally sit at the boundaries of our skill capability can I add two quick things there Ben so just so that was a great explanation but just to hit two points a little bit harder one you might think um all that matters existing error that's why it looked a little bit paradoxical that we also pre antithetical to our modus operandi which is to reduce error but when you remember that for our contradiction error minimizing system we have a really deep temporal model we're managing uncertainty at a big Horizon uh maybe even multi-generational you know where we're thinking about our kids or our kids I mean who knows how deep the generative model actually runs it turns out that that stops developing error minimizing skills and abilities and just hangs out in one Micro Niche because that Micro Niche is going to be upset sooner or a little bit right I mean it's such a great example of where we thought we were in a really set Vector State and then suddenly it gets jostle then all of us go oh my goodness like what do we do we've been bumped um and uh it turns out that the best air minimize will be the responsive to kinds of errors that are Advanced and it hates those errors are digestible so that means not too complex that you can't do anything with it you can't learn anything from it but also not so boring that there's nothing to learn if we hang out if we're sensitive to and we hang out at the edge of our capability then we keep developing new error minimizing abilities which actually sets us up to be good error minimizing systems over the long run so even though we're investing metabolism now we're getting we're setting ourselves up to be able to manage you know especially in the deep end of the pool Black Swan funds manage uncertainty that we're not going to be able to predict the really unpredictable unpredictable um that comes from hanging out at this Edge so part of our curiosity and playfulness and creativity are going to be about us making and digesting novel slopes of volatility I think that's one that's one really great answer for why you get playfulness and curiosity out of this and creativity I'll just drop one more and we don't have to get into it too deeply here but here's one other one that we're thinking of there's something really special in the creation of art especially in a public sphere that I think is so interesting and that somebody needs to be sort of looking at this um we there's actually a new collection coming out in philosophical Trends B um on Art and predictive processing which I think you should check out if you're interested in these topics um but the idea that we've been thinking about is there are ways that we can bring our generative model out and put it into a public sphere you can take something which is typically on the inside and you can put it outside and then you can have other error minimizing systems look at it and Fiddle around with it and then we can re-imbibe it um I mean we're doing this all

the time I mean that's what language allows us to do and what writing allows us to do think about maths you're bringing a part of your predictive understanding you're laying it out and other predictive agents can fiddle around with it and then you can take it back in as part of your updating your own model and I don't know how much more I can say here I just think this is a kind of horizon but I think there's a really juicy thing to say here about where real art is like Tolstoy maybe Leo Tostai was already on this where you put something of the Creator in the creation and then other people receive that and then they can they can um they can do something else with it and then you're able to sort of take it back in all under the guise of sort of minimizing volatility or understanding volatility maybe that was a bit deep but uh no that was really that was pretty good yeah yeah um I just didn't so in the chat I just dropped a link to um um yeah so somebody just asked about papers related to hanging out at the edge of our capabilities I'll do that right now uh yeah sure awesome papers on that uh Mark's got a few I already dropped a link in there to an aeon article by um some of the old expect project crew including Mark Kate Nave George Dean and Andy Clark on the value of uncertainty which is I think it's a really good start point for some of these questions um just because it's aeon is a kind of non-academic um place where you can you know it's a it's a non-technical introduction and there's a wonderful example in that paper of a guy named I think it's Max Hawkins is that right Mark Max who um he's a kind of tech guy and he realized he was getting kind of bored with his life he inadvertently trapped himself in a dark room because his life had become wrote and scheduled and systematic he realized that he would be kind of incredibly easy to kidnap because he was in the same place at the same time every single day and to get he took drastic action to break out of that dark room so he he kind of introduced a randomization algorithm that would it would kind of choose for him where he was going to eat where he was going to go which shows he was going to attend and this example is a really nice centerpiece in this article so and I think yeah I think this is a nice place to start because it's also grounded in discussions about epistemic actions as well because there's also you know as cool as play and art and creativity are there's also really really good reasons that uncertainty is valuable as well right because while we might want to exploit all of the opportunities for kind of nourishment and valuable predictionary minimization in our environment there are of course going to be times where um those those opportunities are exhausted and we need to go and explore different ones and I think there's some there's there's some stuff in there as well on epistemic actions within the context of navigation as well so sometimes we're going to have to we're going to have to temporarily move further away from our Target in order to minimize

our uncertainty about where we're going so there's lots and lots of stuff on the value of uncertainty um and it kind of it's very much at the Forefront of some of the really interesting philosophical applications of the framework for sure okay yeah by all means yeah we've got time you just said something I know you have a question that is but with the value of uncertainty it's just that I'm thinking because I've been working a lot with The Magicians and there's a way of using uncertainty as entertainment and it's very difficult to understand it's like you know as a concept for entertainment why would you like to be in a place where you cannot you want to predict but you cannot and you still want to participate in this activity so for me magic shows it's gonna break in a little bit yeah I think my I spoke to um I forget the person's name but there was a Escape meeting here at Sussex and I spoke to one of the guys on Xscape who had been working on Magic um I don't know if you know who I'm talking about but I think they have a book on on Magic um and I think that's really cool to think about from a predictive process in perspective the idea that magic is just these wonderful kind of violations of expectations in a certain sense I think predictive processing provides a nice really intuitive framework for thinking about those kinds of things one one little point there uh Regina just about again if it feels paradoxical just to sort of um complexify your view remember that the this is always happening in a hierarchical system so just because you have errors at one level of the system doesn't mean you're going to have sort of critical cascading unbearable errors at higher levels so that's why it's fun to go to horror movies so I dropped the paper here also we have a new paper coming out on on horror movies I'm pretty interested in this um but it's the same as the it's the same as going to the magic show in some way now the reason why those errors are fun is because we're safe in a theater um we're with our friends we have lots of sugar in our system from eating popcorn and Candy um we can control the amount of scary by by you know covering our eyes or like there's lots of control here and yet we're getting media that is that's pinging all of our evolutionarily ancient error tracking systems so that we're getting jumps in the physiology as if there's a bunch of volatility that we need to manage and yet we're in a completely safe space which is really fun for our kind of system because we're hierarchy LED predictive systems the high level here isn't jeopardized it knows I'm in a theater but the low level stuff is still registering all sorts of little volatilities but of course they get squashed they don't Cascade all the way up um or or if you're the kind of person where they do tend to cast all the way up then you're also the kind of person who doesn't like going to horror movies because it kind of scared like you go home and think oh goodness maybe ghosts are real and maybe I live with them you wouldn't be going there then

so with magic it's the same sort of thing um we know to a certain against so with errors I don't know if you've ever seen um who does the really extreme magic David where he like was in a block yeah David Blaine have you ever seen him he goes to Haiti where you have a community that really believes in Magic and he does some magic and he gets into trouble like suddenly all the young men are like oh no no no no that's that's no bueno that that you're doing this and then he had to the camera pans back and he goes no no hold on I'll show you here I'll show you how it's done it was just a trick I'll just show you outside it's not real magic because there they're not having fun with it anymore they're like no that's like not a good thing to do you know so it just goes to show that like our our engagement with magic is mostly a sort of scary play yeah I think one thing I just kind of sorry about the seagulls I don't know if you can hear those there are seagulls going crazy just outside my window um one thing I would add is I think um another aspect to this this part of the framework wherein you have this value of prediction error kind of epistemic and affective emotional value of prediction error it's one of the kind of key elements in this notion of uh agency get in active inference that I think is really it's gonna be really important in mind as you go through the further weeks of this course as well so what you know like I said in the lecture it's not necessarily speaking to a quite the metaphysical question of free will but it's certainly moving us away from this picture of the agent as a kind of automaton that's just following rules There's real space here for kind of individuality and creativity as well okay Darius has been waiting a little while so um let's go over to him happily patience happily um yeah I mean this is this question I guess is deriving from my thoughts about the meeting of the the high road and the low road some of the recent work that's been coming out on Maxwell ramseed an actual conversation I had with him which is that the generative model of the states so my question given that give it an tall is regarding the notion of affordances I think it's natural that as cognitive scientists and philosophers we take the perspective intuitively of the agent in the agent to reading a relationship and how the agent is in the business of reducing prediction error I was just wondering whether as philosophers um you and Mark have thought about what it would what is it like out for the whole system to be in the business of producing its uh prediction error so I'm interested in flow States um and so for example my thinking is is that in the canonical example of Thursday let's say a rock climber the the climbing ball is also in the business of losing its prediction error as to afford the opportunity for it to be exploited stealth or themselves are in the business of reducing uh prediction now so I'm wondering whether this expansion of affordances is bi-directional traditional infinitely directions better that's an amazing request so um you uh the

Revenge the Axel constant that somebody uh someone else but I spoke to Maxwell ramstead last week um yeah yeah there's a there's a paper on uh I'll find it in a second on uh Nishka instruction um and um affordances I think it's the Symmetry between uh a niche and an agent um very much in like the same vein as you were just talking about there this idea that it's not just uh there's not just this unidirectional kind of fit between the Asian and the environment but the even or the niche is actually modeling generative model of the agent as well implicitly um so I think the best I can do that direction of some really interesting work on that but it's um certainly it's not it's not something that plays a central role in my thinking recently um yeah so we have the participants Zoom chat a variational approach to Niche construction yeah I don't know if uh if Mark has anything to add or anybody else which we're thinking yeah yeah can I just check one thing you asked I I don't know how much I need to add here but um did you think that the wall was reducing free energy relative to the climber did you say that that's right so my so I don't know how that can that be the case can that be the case the wall the wall is a thing is maintaining itself and I mean relative brain self-evidence thing right yeah but but it's not really a duet of one here it's not like it has a generative model that's deep where it's modeling the climber modeling it like the same way as tango dancers do the Tango the a Tango Duo are are are having Cooperative flow States because each movement opens a Vista of possible error minimizing opportunities for the partner who in turn opens of this set of partner yeah and backwards and forwards into this ever opening expanse of affordance capabilities but an inanimate object in an agent don't really have that same Dynamic so I was wondering what you were thinking there so my thinking was is that this kind of bleeds into the notion of fluidity of affordances in the kind of gibsonian sense and the fluidity of Concepts so I agree in this in the sense of physical self-evidencing it needs to model the rain it needs to model the physical environment so as to maintain itself but what about as it's a it's a it's because so a sheer rock face is not a climbing wall and that's because it doesn't self-evidence as a climbing wall it doesn't offer the affordances to be climbed that's why I was kind of referring to which is that affordances change based on the on the agent Arena relationship right your cause affordances change whether it's working or broken down similarly the climbing will changes whether it's got footholds handholds or whether it's just a sheer rock face that was my thinking it seems slightly spooky it seems slightly spooky to me that we'd think of the Wireless self-evidising in that way I think I think the language that we tend to use because Julian kivocene Eric reidfeld um you know the Bloomberg and folks often use the idea of affordances and Fields of affordances in the active inference way um the the radical end of the

pool there is typically that the affordances don't only happen on the agent side but rather happen from a dynamic of the changing volatile environment and the generative model of the agent and that their their collaborating in Dynamic ongoing ways such that the affordances are emergent between them that's certainly that certainly seems right to me the the idea to push it a little bit further and think about the wall self-evidencing it's a it's a I think it's a step too spooky for me um I think I would I would feel safer safer a little bit closer to thinking about yeah the affordances are changing relative to the Dynamics of the wall um but I don't know what it would mean for the wall to be evidencing the climber or itself yeah there's uh just to kind of that's instead a ton of thoughts well a little longer on um affordances because there's this really nice distinction in their work between uh what they call well I'm not sure if they I think I think that the term landscape of affordances is a little bit older but they make this distinction between a landscape of affordances and a field of affordances and I think the reason this is it's really important to get on the table is because it this distinction is kind of directly targeting the dynamic shifting very affective nature of affordances so on the one hand you have this landscape of affordances that is relatively static so so right now my landscape of affordances is is Brighton um and that's the landscape of affordances in that sense it's not really going to change very rapidly unless I kind of jump on a plane and go somewhere else but the field of affordances has this thoroughly kind of normative affective character to it so depending on my internal State at any given time depending on my expectations or my desires say my field of affordances is going to shift very very rapidly and it's so not just not just predicated on my internal state but also kind of contextual cues as well so if something were to happen in the environment that was afforded very high Precision waiting then that's likely to shift my feel affordances significantly I think that the reason that I'm kind of emphasizing this to Darius's point is and and kind of building on what Mark said it it doesn't seem like this distinction at least would apply from the perspective of the perspective of the wall say because this kind of affectivity and normativity just isn't a feature of the world's experience of the world and also to kind of run with that the I mean affordances perhaps you could say a bit more about how you would kind of think about this Darius because affordances are kind of opportunities for actions okay so it's they're they're kind of uh I I like that there's some debate about this but I like to think of them as like a relational property between of like this you have the skilled capabilities of an embodied agent and then you have some feature of the environment so I think when you're thinking in from the perspective of the wall you have the features of the environment because you have the embodied characteristics of the agent are

kind of playing that role but it's not clear to me what the the kind of skilled capabilities of the wall is it so yeah if you could just say a bit more about that yeah I mean it may be the case as you pointed out that actually the the technical term affordance as being obtuse fact has misplaced her okay so I can see I again this is a lot of this I've been formulating having read this very new paper by um Ramsey and colleagues on Bayesian mechanics of physics of them by beliefs where he really Rams home this point that the system the generative model is the system of the kind of statistic uh stochastic differential equations across the state space so across the mark of blanca across the particle with its Market blanket and the external States and that is the system which is in that system itself is in the business of reducing free energy so it's not just the active inference agent yeah the whole system yeah which is embedded in the whole system of other things and so that was my general thinking is that yeah just on the prerequisite that these systems exist just on that aprily Axiom there has to be some kind of self evidencing of that system itself so maybe affordances is the wrong word I yeah yeah that might be the trick right there as soon as you provoke affordance yeah you're provoking phenomenology and so now we're not talking just about statistical variations now we're talking about lived experience that might be that might be the the linchpin right there I think I think that is right yeah and I and I would say I think Darius you're absolutely on the right track and you're in I mean there is this ambiguity about affordances but aside from that I think you're absolutely right I mean it's I it's just nonsensical to think about an agent's absent an environment under active inference you have to it is this agent environment system in its totality that's minimizing free energy they have to be they have to be taken together as one and I know Abel you've had your hand up for a little while though do you want to is there something you wanted to add yep uh do you hear me well we do I do anyway okay um about the references World Arena foreign season um basically the symmetry of hydroxine environment is a property of active in France well of the financial principle so if you buy a friendly principle and you also buy that it entails that agent have affordensies or any proximal notion then you also buy that rolls as uh forenses so there are basically three position you can have on that one is that active inference is correct well a [__] affordances and they do self-villiancing which I would not go with another uh option is that you have basically um Devol in the details so for example Maybe the adjunct environment is not the wall it's something that is broader like Gaia system whatever and it just happens it includes the wall but it is not the wall and it does do self-balancing and the third is that active inference is wrong uh we don't have affordances based on whatever is the follow presentation of active in France and I would

go in that direction of this phenology phenomenology stuff so we have a I I would say philosophical evidence in the sense that they are denying this would lead to nonsense but the information is based on observation and some system do observing much more actively than others and maybe it's a good thing to have in your formalism I'd say uh so um you have maybe a stronger case for uh the way uh if are constructed in the quantum formulation of the fep if there is such a thing which like Chris Fields someone who we should Baseline believe claims there is because then you have formalization which is not in terms of Dynamics per C so causal constraints but observation and indirectly phenomenology maybe you have something that is stronger there but right now the uh link oh and yeah sorry there is actually a bunch of people who have a quite good hold on science which are the SE based on the ecological activity and commission on one hand and between cognition and the metabolism the constitution of the living thing that cognites on the other and one of their definition of agency cognition pick one entails interactional asymmetry I think is the word that maybe something we would need to translate into active inference well into the fep for active inference to have a strong grip over this affordance thing until then we have a kind of hand waving way it relates it to more conceptual approaches uh that are related to an active and ecological approach to psychology and um yeah the formalism lags basically behind the concept for now sorry for the long talk no that was great much appreciated this um anybody else want to come in on uh question of affordances because I know um affordance has played a fairly substantial role in the lecture so anybody the questions related to affordances or you know action perception affordances then Now's the Time yeah I do have one but it's not well structured yeah I've been thinking about for the last months because I I started to deepen the artificial intelligence field up coming from a Humanities background and philosophical background so I'm mixing many things in this period mainly following the the active inference as a as a Crosser between many disciplines that and this is very beautiful about the framework and I was super curious about the um the question proposed by Darius and I was then thinking about now we're actually facing a a new environment and a new landscape in which we're like interacting with the term comes the next turn actually generating over revolutionary time and synchronizing a logical alien agencies operating within our landscape and field of performances um so I think there's a lot of really interesting work to potentially be done there I will take the opportunity to shamelessly plug a pre-print that Mark and I have released recently where um we one thing so one thing that I'm really interested in that I can kind of speak with some confidence on is the fact that I I do think there's a point at which the theory of affordances or the language of

affordances starts to break down legit so the preprint that I'll show around is looking at and um ambient technology specifically so this is technology that you the whole kind of impetus behind its design and conceptualization is that the user doesn't have to actually do anything so it precedes the user pragmatically and epistemically it knows what kinds of things you want it to do and it just does them in the background and it works by shifting kind of it it subtly shifts the material environment such that it impacts your field of affordances in real time and and the argument that we make in that paper is essentially that the affordancers approach doesn't really work for this and that we need to kind of think again so um yeah I mean I think that that might be it but like how to conceptualize generative AI for example under the affordances framework I'm not sure I don't know anybody that's done any work on that but it would be it would certainly make a really cool project you are making it better and right now when was that you are making it right now I know because you sent me the paper to review oh yeah thank you thank you so um we were looking at we so we uh or early stages of a paper on the role of um generative AI in classrooms um so from the perspective of thinking about what kinds of affordances generative AI represent in a in a learning environment um particularly we're kind of applying active inference to classroom design and looking at it from the different different uh the angles of different um educational and pedagogical theories um but it's um very early today with us yeah this is I suppose it probably depends on your view of generative in general I mean it depends whether you would consider it a real cognitive agent or not I think I think some people are more inclined to think of it as you know this thing this is a thing that has agency it has real understanding it has like um you know and other people are maybe less inclined to think of it in those terms and I think that might be that might bear significantly on on how you think of it yeah that's super cool that basically the last project you mentioned is very very in line with my PhD project right now because my main PhD project is our European findings and it's about the AI for personalized education and I'm trying to follow it through the the active in French framework from a multi-level perspective so yeah very cool yeah we should we should probably talk more about that okay uh Darius yeah I wanted to ask I mean this may be directed more marks I know this is kind of his work in terms of um slopes of uncertainty and about doing better than expected at reducing prediction error over time I was wondering kind of how the architecture of that is built into the into the particle into the into the genitive model because we have the kind of for me there's I don't know if it's an hour and out sort of conflict but we have these kind of implicit priors that we're going to fulfill certain expectations or minimize prediction now regarding certain things right so homeostatic

priors or happiness well-being whatever it is but then your claim is that we have a higher order beliefs the higher order predictions that over and above that I also need to become I also need to do well better than expected a reducing prediction error over time so is there a it's still quite fuzzy in my mind but is there a kind of potential conflict there between the general priors that the system has and then go actually in a sense violating those Expectations by going over and above them which itself constitutes an expectation how do you kind of resolve that tension yeah that's interesting um but this is very much in Ben's wheelhouse too um Ben and I have been working on slopey stuff for uh almost as long as I've been doing active charge what was his original paper called we'll get it we'll get it up they do a really good job of showing technically where this sits within um uh the parametric model um Laura served Smith's paper on metacognition also does this by showing you have these you have these depths of modeling where models above are modeling models below okay and optimizing over those models below so we have some good computational backbone for thinking um for not only thinking that this is the case but beginning to express how it does the work it does um so let's let's leave that for digging into it though technically on sort of on our own let me say sort of a at a higher more abstract level still hopefully it's useful um it's not weird to think that this kind of anticipatory system is not only making predictions about the world but it's also um part of what it's predicting is how fast or slow how efficient it is in particular contexts at resolving certain kinds of errors that that's a perfectly fine thing to think that we're also predicting slopes of Engagement and then and then all we're saying then is system also pays attention to when those expectations are breached it's learning from those breaches that's that should just be the bread and butter for what the system does anyway so Precision is a second order is a second order process in much the same way Precision is about how well uh how how reliable are lower level predictions um and then using that um that amount to toggle how impactful either errors or predictions are so we've already got baked into the system right from early days this idea that the system is not only making predictions but monitoring its own predictive processing regimes and then toggling uh based on how reliable those those substreams are um all we're adding here is um you know when we first when we first thought about those mechanisms one thing that can happen this is just good for everybody who's interested in active inference because I bump into this with my students all the time when we say something like Precision waiting a tendency to think it's the same so we go to find what is precision like where's the biological instead or I guess Precision waiting when we actually get into a bio system and we look for these things the truth is precision is going to be weighted in

lots of different ways I mean that's the real Frontier of This research is to actually find how these things are instantiated and the answer is going to be multifarious I mean it's going to be you're going to have Precision adjustments happening throughout the system in lots of ways it could be synchrony and asynchronies and desynchronies between systems it can be neuromodulatory chemicals it can be structural structural um uh shapes within the brain I mean Precision is going to be set in in lots of different ways so uh uh all all that we're pointing out here is is that one of the ways that Precision is being set one of the ways that the system is tracking how efficient it is and then upping or lowering the amount of impact um error signals or predictions have is that it's happening in an embodied way so we've looked over to the effect Active search and and lo and behold there are all these signatures that we were looking for for this kind of for this part of the Machinery so um yeah so not weird that the system is tracking its own regularities and adjusting that was baked in right from the beginning how it does that that's one of the Frontiers it's going to happen in lots of different ways lo and behold affect the Dynamics the shoe fits they look like they do that stuff and then you bring it to the lab and you go back to like reward prediction error research and sure enough neuromodulatory chemicals reward systems are tuning affective Dynamics relative to better than and worse than slopes of uncertainty management that's exactly what we would expect given the computational model um so then it's an easy it was an easy next step to start saying well look there's one of the way it's a Precision that there's one of the ways that after effects [Laughter] system now just notice there and now the only thing the affect the system is doing we never want to say we never sort of we're careful not to be reductionist to say oh affect is always aerodynamics I don't think that's right I think it's that aerodynamics are expressed in part effectively and they have this impact on the system um that we can that we that we've known about for a long time even from just a reward prediction error um literature does that help or was that a bit is that okay yeah yeah yeah uh yeah uh just to just to add two two things really quickly there um because I know time is kind of catching up with this uh I would emphasize as well like just as a kind of General point the one of the things I really like about the work on aerodynamics and affect is it has this nice kind of broad capture of the kinds of affectivity that agents can experience so we're not just when we talk about affectivity we're not just talking about full-blooded emotions or even just moods but um we're talking about what is it one of the things I kind of Drew attention to in the lecture was Matthew ratcliffe's work on existential feelings which I think is just such a um that the connection between active inference and phenomenology that you find in some of this work is just insanely powerful and it's one of the

things that really drew me into the framework um the second thing just build one thing that I don't think I mentioned in the lecture is uh active inferences it has been described as a quintessentially metacognitive framework so you have kind of built into the architecture you have expectations over expected so you have kind of predictions about predictions and this has proved to um just you know again just phenomenally useful for thinking about certain aspects of phenomenology as well so um yeah I think that these are these are like real strengths of the framework um I don't know how we're doing for time are we are we strictly limited on or can we we are not as far as I know Danielle well um I can probably do I'm definitely do another 15 um if anybody has any more questions and people I should say as well we're not that doesn't mean that we're all held captive um to my time frame so if people if people do need to leave within the next five or ten minutes that's absolutely fine but um I'm certainly happy to to carry on if anybody has any more questions or comments um if if no one else has got any I would quite like to go back to uh earlier in the discussion you were talking about the value of uncertainty um because uh in organizations in which I'm working uh we often have conversations about uncertainty and what we find is over time there tends to be a drift towards risk aversion so we're working with a large uh International construction company at the minute and that they used to have a culture in which innovation um uh was uh was quite well embedded and and over time it's kind of drifted towards very very risk-averse culture where actually rather than learning through the process and being able to tolerate uncertainty you know around opportunities and you know what can be accomplished and that kind of the valence that comes with uh the intrinsic reward I suppose of being able to reduce uncertainty about capabilities to a situation where they're trying to anticipate all of the risks up front before they even get involved in the project so you know there's some kind of valence switch market and you mentioned um David Blaine earlier you know and something about the context you know there was a switch in valence from yeah this is this is a kind of a play this is a safe thing to do to actually know this is you know this has real consequences so I'm just kind of I'd like to explore that a little bit more actually if anyone's got kind of any insights or papers or comments yeah so I think that's a really interesting question actually and I I think um so it seems like one of the things one of the things we've been talking about is this connection between uncertainty prediction over minimization and affectivity and individual agents and it sounds like what you're asking is like how does that translate to uh like collectives so yeah one of the things about active inference that we're going to see in subsequent weeks is how it scales up to kind of larger systems like companies or you know groups that are trying to

achieve some kind of shared goal um how do you how does the value of uncertainty translate onto like how is it scalable in that sense right is that does that be a fair kind of yeah I I guess so um I I guess it's um I mean I so uncertainty often when it's talked about uncertainties talked about as risk which is you know I suppose is you know predictions or anticipation of uh outcomes that we don't want right whereas there's also positive uncertainty right you know which is you know I suppose you know simply stated opportunities you know through curiosity you know what might we be able to achieve you know this kind of novelty seeking I guess that you know new opportunities that we haven't exploited um and they're just you know in the kind of organizational systems uh you know I think there are various pressures that cause it but but over time you see these cultural shifts towards uh you know a very very kind of risk-averse you know where you know where the valence is obviously quite negative you know you know quite an unpleasant feeling for people and yeah yeah this is good this this is right at the heart of my current work um I'm really interested so you know we just had a big stint where active inference models were starting to be used in computational Psychiatry especially for pathological disorders so addiction depression disassociative disorders OCD PTSD um it's a really it's quite a it's a quite um a sexy framework for thinking about some of the ways the cognitive system breaks down in the sort of new move right now I mean we have a collection coming out right now in Neuroscience of Consciousness is to think about these things in terms of if we are predictive systems and we have a sufficiently rich model of how that system works in a particular Niche what sorts of what sorts of um what sorts of ways can we intervene on that system in order to have positive outcomes rather than just modeling what the negative outcomes are and I hear that a lot in what you're saying now so I'm just going to drop one link for you there this is Casper hasp again um wrote a little a very small paper with a nice little model called sophisticated affective inference gives it a little bot that tends towards catastrophe catastrophizing the future if it's given if it's given like fun medium fun medium fun dangerous over time uh it tends to it tends to expect the dangerous one um uh like after 10 000 iterations it basically lives in the worst possible scenario which is a nice little this is my wife this is this is most people today you know so the question is um I want to know given them all model why does that happen and what can we be doing to intervene and um part of the answer is going to be we need to become more tolerant of uncertainty that's one of the things that the system can be better or worse at so this just takes the computational modeling and then looks to things that we already know about emotional regulation that that's that's part of the story um so um we've done a little bit of work on this with our predictive dynamics of happiness and

well-being and Ryan Smith is definitely doing work on this with active inference and well-being um so definitely check that out if you're if you're interested here I'll just flag one interesting thing here that relates just to what we were just talking about about layers of modeling one way you gotta let's say two ways there's two ways that a system and it's probably gonna be lots but the two that come to mind that A system can become more tolerant uncertainty is one exposure so this is why exposure therapy might be useful for our kind of a system you want to expose the system to volatility at the lower and middle levels of the hierarchy and have it turn out okay so that this one of the things that the system can come to predict is not only you know particular outcomes but it can also predict how much error is involved in particular outcomes so for instance when I used to give a pro talk I was always really nervous um and I don't know if that ever really went away but I've had so much exposure to the anxiety of giving a professional talk that basically I don't notice it anymore but if you were to ask me at the beginning of my talk Mark right now what's your phenomenology and I sort of meditatively looked I'd probably say yeah I'm nervous but if you hadn't have said that and you was like hey what's up I've been like oh yeah I'm great like it's all good I don't even notice it anymore that's because the system knows that I have errors but as soon as I start talking they drop away and because I know the arc that evens error in the system it becomes non-newsworthy it's no longer interesting volatility to be tracking that's just from exposure okay so what's happening is you're having errors at a lower middle level that a higher level is now modeling saying when we come here we should expect this Arc of error same thing when you work out like real gym routes they can feel good the legal system right but at the higher level it's now learned not only that that has a natural Arc but that that's a good sign at a higher level so now you're getting a positive you're getting positive prediction error slope high and negative prediction air slope low okay so one is exposure two you can model your own responses this is something Ben and I work on with um horror movies um uh you can you can take an active role in mindfully observing your own reactions to volatility and um this comes up in our paper on horror we've already dropped the link today if you want to check it out something really special happens when the system models its own reaction to volatility it starts to learn that um even reactions to volatility uh don't need to be um they don't need to be compounded up into dangers that it's okay it's okay to be uncertain at certain levels of the system so how that translates into business I'm not sure other than I know tolerance to uncertainty is a marker of business success and um I suspect the framework in a fit lots of mindfulness mindfulness work about learning to tolerate uncertainty is there anything about

um the role of language in kind of modulating those metacognitive oh it's good if I'm I Was preparing an answer on that specifically and I was nervously waiting for an opportunity like meta in terms of evolutionary cognitive archeology we don't know when language emerged but we do know that's uh let's say an encouraging semantic system so you can build things in language and you can expect things in the world to correspond to things in language so for example if you that's abstract as I say it's abstract but you can for example have a self-identity French I'm a French and I communicate with the expectation that are underneath with other people and that means we're communicate our nose so so we have specific expectation of what we will do and those expectations they become embedded in specific symbolic markers that are embedded in language and uh something that uh like this thing that it was strongly evoked in my mind when you talked of three words to answer that is that um as far as documented history goes uh Europeans well indo-europeans they are pretty strong on the idea that things of a nature and they act lawfully and their nature and their laws they somehow correspond to linguistic categories so I can say stuff like chickens quack and that's a proper explanation of what are chicken and why they quack and if you look in Chinese philosophy as far as written history goes you have a major Accent on way way so um a poor translation would be uh effortless action closer translation would be uh action without action so it's a form it's something that is quite close to the phenomenology of flow in that you observe yourself doing things and you do not apply a conscious effort to the flow of what you're doing and that looks like something that is uh closer to the phenomenology would expect if you apply you know recursive proactive condition with less or less heavy or more effectively integrated symbolic occurring like use of language as something that you actually expect to be meaningful and to constraints from your actions and marginally I'd expect very strongly do you expect that linguistic categories will map and what you build with it so I don't know self-identities plans whatever will map cleanly onto What You observe you will be very very unsure about things because that does not happen usually and yeah um besides uh Grant discourse on interpreter preference traits which are usually not that constructive I'd yeah sorry I got lost because I did not I that's a speech tonight can I let me just say one thing before before we move off this point uh a simple way that language might help is by invoking cognitive flexibility so um oftentimes we think about the management of error being either updating predictions or acting on the world those are the that's the dyad usually what it overlooks is the third one which is we can also manage volatility by redeploying Precision in a better way so rather than just updating your model to fit the world or updating the world to fit your model you can just

change the set of what matters so that you're not you're not really updating the model and you're not really changing the world you're just changing the problem landscape and that's something language allows us to do it's maybe one of the one of the really amazing things that language allows us to do is we can use language to bootstrap that kind of precision adjustment so if the train doesn't come on time we both notice it okay and we feel bad that's perceptual updating we might go and get a taxi that's active updating but I might just say to you um wow look isn't that isn't it lovely that now we get a little bit more time to read our book or to continue our conversation or let's finish our coffee with a little bit of ease I mean we get an extra 20 minutes now even just saying that redeploys Precision over the problem space now the train being late isn't volatility in the system the train being late is signaling to the system that a better than expected slope has been achieved isn't that so interesting the exact same occurrence is either undigestible volatility right like you're gonna be late oh my goodness or it's an opportunity for an improvement in the system which is now you get more time with the person that you're um taking the train with and that was just a matter of linguistic um perturbation of the way that Precision is being deployed so I if you're looking at if you wanted to dig into the research here I would look up cognitive flexibility and language or coaching cognitive flexibility is such a good point here so the uh you have a paper by any clerk on uh this specific and current role of language and something it entails is that basically you got like if the active influence picture is correct you will flexibly predict to flow in the uh interceptive proprioceptive exteroceptive not even a thing space and that is that is what Builder integrating experience integrated you know cognition and language it adds another layer of complexity if you can just talk to yourself in your head that is another dimension that you can predict and that can be coherent or incoherent with uh you won't and so that is an extremely powerful and current system because I just have to like tell a plant to myself and boom that uh pushes me nudges me nicely for the plan but then you can have a different level of meta expectation over how language corresponds to reality and I'd say a very very strong factor of you know a version of coincidentally is whether you expect your plans or your linguistic structure to nicely map them to reality because it will not happen but if you expect strongly that it will be the case you will have to make it happen somehow and here you will adapt rigid strategies and you will like lose flexibility but also language and sorry for flexibility to be the case in the first place so it's about coupling between dimension of cognition yeah and I would just uh just kind of risk and danger are two things that I've been really interested in across a kind of variety of different contexts and I I would say some of the things that have been said

like the kind of how you can influence the system by externalizing things through language and the different strategies that systems whatever scale they're on have minimizing prediction error um minimizing free energy but also I would just emphasize the importance of like external context here as well so even in the case of like a collective like a business you still have to take into account the environment in which the business is operating and I I've been really interested in a particularly extreme example of context um I used to do still do some work in philosophy of sport and I'd be really interested in dangerous sports and why there seems to be this insane contextual effect where within a very kind of narrow contextual band of sporting practice people seem willing to take on risks that outside of that context would just be completely insane um and I think this is something that's probably going to come up again again in uh you know subsequent weeks as well here when we talk about um the way that expectations on an individual or a collective basis can be set but through other Minds uh like patient so what somebody else expect myself to do and and so on and so on uh Darius you've had your hand up a while do you want to jump in yes please I'm not sure this is the sort of grand synthesis of any of this but um hello was mentioned as I said that's the kind of at topic of mine and something that we know about flow and this way it's integrated into this idea of language and agency is that at least in its original conception of success mahali and the sort of qualitative experience associated with flow you would find stuff like not only the dilation of time but also the reduction in self-consciousness and now this is sort of picking up a lot of work that happens at UCL and people like Jeremy Skipper how integrated languages into the sense of self and it seems to me and I you know this kind of shooting from the hip but all of these very deep-rooted I guess they're kind of psycho Technologies the self language seem to dissipate at this Goldilocks zone at this point of at the at this edge of criticality at this Flow State which makes me think if the flow state is a phenomenological offshoot of being or you know optimally reducing prediction error what is the kind of role of the concept of the self or linguistic functions because it seems to me that that becomes a regulatory function when that optimality is reduced when soft starts going wrong um so I think you can also think about this in the kind of hydrogarian or Dreyfus sense of just opening a door you only start to represent the door in yourself you only give it the linguistic object of a door when you can't open it when you open the door standard there is no representation going there I mean you could even argue there's no quality out there so I'm wondering how deep that runs and what the kind of what we can say about the role of agency selfhood maybe even consciousness um because they don't seem to be that prevalent at least from my understanding when we are at the edge of

criticality the the suggestion you're making there is um it sounds like what you're saying that these thing you call them psycho Technologies which I really like uh you know like language and uh and and self-mod like certain aspects of self-modeling they are kind of scaffolding uh or like a ladder that you then kind of gets kicked away once you reach a certain kind of edge of criticality like where the performance becomes you're performing at such a level that you just don't need those those scaffoldings yeah or or the self-regulatory mechanisms that we harness when we're not in flow are linguistic or agentic in essence and once yes once you saw you you don't need that when you're at the sort of yes when you're at this edge of criticality um and I you know I only postulate that because I want I'm only I'm only thinking how deep does that go because I know obviously the the active inference framework is trying to tackle in some ways the hard problem as well and there are some flotations of the idea at least within the flow Community about the kind of I don't want to say elimination but the moderation the modulation of qualia on the flow States so could that could we also consider that to be an a construct um which happens when we're not necessarily optimally you know reducing prediction error yeah so a couple of things I so I'm certainly not the person to start talking about Consciousness so I won't um but I would just um I one of some of the readings that I mentioned in the lecture might be interesting for you on this topic so if we're talking about certain structures that seem very very pervasive in our phenomenology and how those structures can sometimes kind of dissolve away and how that might be accounted for under a active inference framework there's some really interesting work on psychedelics by George Dean and some of George Dean's co-conspirators I know I know Mark and Sam Wilkinson worked on a paper with George so George Dean's done some work on kind of ego dissolution and certain kinds of experiences related to selfhood um kind of and I kind of don't want to overstep the mark either but I think they kind of come they fall generally under this kind of Rebus model of um psychedelic efficacy so I don't know if people are familiar with this but this kind of finds a natural articulation through predictive processing and hierarchical self-modeling and and that kind of stuff I also wanted to really quickly flag on this topic I'll find the paper and I'll put it up on the coder because I don't think I'll be able to find it quickly enough now but there was a paper that came out very recently um that showed the efficacy of um internal self-talk on sports performance so there were so they did a they did an experiment with cyclists and they found that when you um when cyclists were allowed to engage in kind of constant self-talk with them so like in a monologue in their head their their performance at cycling was like measurably boosted which I thought was really interesting and kind of

relevant here so but it's it's interesting because it kind of just kind of uh it sounds like it might
primaphasia be in tension with what we were saying about flow States as well right so it's if if
the real edge of performance is is a place where these structures are phenomenologic
phenomenology bleed away um it kind of calls into question like to what extent are these
psycho Technologies effective and in what way are they effective and yeah there's like yeah
just a whole kind of universe really interesting questions so if I may uh sell food indirectly
will be the the central topic of the last two sessions my position about it is that it's selfful in a
sense we use all of the time is quite high level of construct that emerged from the ability we
have to compare our activity to a socially embedded mother of our activity so that is
something that is quite specific to humans because of their linguistic ability and or because of
their tendency to um what tomasello calls the share intentionalities for the ability to
collectively Define a neck plants and so basically this is something that enables a very deep
very um profound and robust transmission of cultural knowledge and transmission of norms
this is what enables the Consortium of norms uh uh but it's quite heavy cognitively and it is
predictable that if you're busy comparing what you do to what's another idealized version of
yourself is doing you're gonna invest less Creative Energy in actually doing the thing that you
would if you just you know did a thing so I'd say you have a pretty direct entitlement of the uh
this flow thing from this model yeah yes this is my conclusion sorry um I think so we've run
uh almost 30 minutes over time and I know uh Mark's marks had to go and I'm gonna have to
go very I think I pretty much have to go so um I just wondered if anybody has any final kind
of quick questions or comments maybe we can move towards wrapping it up um I'll I'll make a
few comments but then you you're welcome to leave and other people welcome to stay um if
they would like So yeah thank you yeah yeah well first um just on a logistical note there was a
little bit of lag so I'll encourage everyone who's listening as well as who's participating that we
eventually plan to as with all of our live streams develop the transcript which will be curated
and published and eventually not so far away in the future that'll be enabled with speech
modeling to generate podcast quality audio so the future will be smooth we promise um but a
few really interesting pieces that I that I picked out while I was embodying and listening um
sir Regina you you opened the discussion with the disc with uh surprise and creativity and
indeed it sounds like it's going to be attention how can a framework who's imperative is
bounded surprise as opposed to say maximized reward how can a framework of his imperative
is surprise minimization and bounding be used to generate novelty and I felt like people
provided a lot of really cool answers and one other thing it reminds me of is uh Doug

hofstetter's notion of spec fetishness I don't know how do you pronounce it but s p h e x i s h n e s s and it's the specs wasp and so he says well forget this whole creativity thing because when we talk about creativity it's like some sort of you know Pantheon like Divine status and abuses have to be uh you know called in and so then oh that's not real creativity and this isn't real creativity so instead he says well let's just focus on what would be the opposite and that would be the road procedural following even when there's an opportunity for what we would call creativity and then there's a Continuum so why does somebody paint that painting well why didn't they invent a new genre well why didn't they invent a new media and so all activity is existing within this continuum of creativeness which has many features such as generativity and productivity but also novelty but abounded novelty it's not creative to to knock the chessboard pieces off in a way that there might be an elegant chess move who only somebody in a certain cognitive setting could detect the Aesthetics of but I think that that was awesome um and then the second uh piece that really was was powerful was what Mark Mark's train the I guess the trains don't run on time where Mark is but but that's okay because he was able to rather than treat that as an undigestible surprise that was able to be basically cognitively metabolized into like an opportunity for friendship through language as a social medium and then he connected that to cognitive flexibility and coaching and then I thought about self-talk and our inner monologue inner voice and then there is that that was very um provocative that like we aren't having self-talk potentially not even having a self in the flow manifolds the experience of a cell for the speech self self talk or again potentially even the total actual self as a technology and so then like healthy self-talk would guide us to the flow and be kind of like a self-limiting technology because it would be like used in order to not be needed to be used which is kind of how we would hope Adaptive Technologies would be versus a maladaptive technology would be something that um entrenches its own utilization potentially at the expense of the functionality kind of like Doom scrolling status but that's internal rumination and those are the kinds of things that people have simulated in active inference so those were some really cool pieces that was a great discussion could I just uh just just add something on the topic of creativity that I should have said earlier as well because I think it's really relevant to Regina's question and just the topic of creative and novel behaviors in general and the right of inference I think that whenever we talk about especially artistic creativity there's a tendency even even for people who are kind of well very very well burrowed into active inference and and activism and these kinds of Frameworks from cognitive science I think there's a tendency to think of creativity as as still like it's almost like the last Bastion of internal cognitivist

thinking like you think creativity is something that happens in here and bursts outwards and so how does that happen with this kind of whatever cognitive architecture we're positing but there's really very cool work by uh have to give a shout out to Mike wheeler um who has written some papers on the extended mind and creativity and one thing that's not really come up very much that I didn't get time to talk about in the lecture is this really nice natural marriage between the extended mind and active inference um something that Andy Clark has been working on and I know he's gonna going to be doing a lot more work on it but the claim wheeler makes and some others um I think Joanna zelinska has written and written a little bit about this as well is the fact that creativity itself is just as subject to kind of uh material in but the the the sculpting effects of material environments and socio-cultural environments as everything else is so create creativity is not this kind of romantic with a capital R kind of internal process that kind of bursts forth but it's creative thinking is just as much extended and embedded as everything else and there's some really great examples in in Mike Wheeler's essay um I don't know is anybody familiar with the band Alt J there are British band they were kind of really big a few years ago they won the Mercury music award because they have a super original sound so Alt J have this really quiet tinkling uh yeah they are a great band they're one of my favorites and they they have this really quiet pinky sound that everybody kind of assumed was just the result of some kind of internal genius on the part of the band and as it turns out Alt J originally they they tried to rehearse as a as an indie rock band but they were confined to rehearsing in an apartment block and they kept getting noise complaints and so they ended up developing this quiet tinkly sound because it was the only sounds they could rehearse that wouldn't get them kicked out of their apartment and I think this is a really beautiful example of how like even the most creative processes that seem on the face of it really creative are in fact just a subject to these external kind of this kind of agent environment system that underpins the whole framework so that's something that's going to apply to absolutely everything awesome and just the last question people are welcome to to drop off um the next section that we're heading into I'll be doing the lecture that's going to be in August and it's going to be on Collective Behavior so what would people like to learn or focus on about Collective Behavior anyone who hasn't um spoken or or it can just be a thought question but I haven't prepared the slides at all I'm happy to take suggestions yeah Darius and then anyone else I think what might be interesting is if we're saying that these uh systems are all in the business of producing prediction error of self-evidizing why do we see a diversity of um why do we see a diversity of social cultures of norms of um standards at a global scale why do

we not just see some homogeneous way to reduce picture there which would manifest as a kind of singular culture which which is in the kind of business of self-evidencing that just kind of popped to my mind cool anyone else want to give a thought on Collective Behavior or on any other aspect otherwise it's been great I would like to make a comment of creativity so but uh does anyone like it's romantic if you want to make a comment on the next session I don't want to put that Ben said that creativity is intrinsically very hard to model because uh by definition uh creativity is something that brings about something new let us say and the mathematics we have to describe physics and life they are not very good at new things because the most basic tool you are the basic way to represent a system that literally everyone uses is a state space which is uh the list of all possibilities of a system if you say something is creative you're likely to think that it means it can bring about new possibilities and this intuition it conflicts directly with the very basic use of math to describe it so it is basically the same issues I was referring to earlier when I talked about the comparison between active in France and an activist so let's say uh activity biology inspired model of cognition and a core concept historically in those circles is autopoiesis so the ability of the living to self-create literally this is Greek for self-creation and you are the a lot of drift and conceptualization and ambiguities that were resolved or not resolved and led to the crazies of frame of blah blah blah but today we talk of autonomy rather than autopoiesis and with the final autonomy as uh the property for a system of constraints over the activity of a system to reproduce genitals and because we're talking constraints uh just the ability to influence causary outcomes there is no really a prior that Dynamics in this space would be conservative so we have a possibility for constraints to you know bring about new things and reconfigure themselves and the notion of hydrogen she had used is based on that it's the ability to let's say reconfigure constraints in your environment and always the notion of creativity which would be likely a very close proxy to that but it's something that we do not know how to prevent and most of the mathematics that exist are structurally enabled represents and so yeah that's a big one foreign Michael uh I uh good to be here sorry I was late uh what comes to mind for me at the collective you know I mentioned units of collective life what is it a wee is it a we Affair and uh you know but you know what what are the what are the units of we and any thoughts or insights about that uh the the stages of the emergence of formation into a collective uh from a me to a weed for example or um uh because I had a one that was on the tip of my tongue and it just escaped me um you know this idea of precision that um that I think it was Mark Miller was talking about or Matt um saying um you know uh at the uh if there's actions what might be some of the most well understood

limiting beliefs about when we use language of collected that keeps us repeating the same blind spots um uh and how might we say differently about Collective so that we we escape those predispositions make sense yeah they'll do what I can do in the solo lecture and I'll I'll look forward to the conversation where we can unpack it yeah exciting thank you for doing it I'm looking looking forward to it yeah it's awesome I'm Nathan or David you want to add anything all right then thank you all hope everyone is um enjoying the course so far thanks aal again for for coordinating it and to Ben for this great section now you can put on your student hat again so and three of organizing thank you see you all next time bye thank you cheers thanks everyone